

HARSHA KUMAR M R

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Professional Summary

Final-year Computer Science Engineering student (CGPA: 9.2/10) specializing in full-stack development, machine learning, and cybersecurity. Published researcher with expertise in building secure, scalable, and intelligent systems using MERN, Python, and deep learning. Experienced in production deployment, threat detection, and performance optimization. Hackathon finalist with proven leadership and teamwork in competitive technical challenges.

Technical Skills

Programming Languages: Python, Java, C, C++, JavaScript, SQL

Web Development: MERN Stack (MongoDB, Express.js, React.js, Node.js), RESTful APIs, FastAPI, JWT Authentication

Machine Learning & AI: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Deep Learning, CNN, SHAP, Computer Vision

Databases: MongoDB, MySQL, Database Design, Data Modeling

Frontend: HTML5, CSS3, Tailwind CSS, Responsive Design

Tools: Git, GitHub, VS Code, Postman, bcrypt

Core Competencies: Software Engineering, Cybersecurity, DSA, Algorithms, API Design, Agile Development

Education

Global Academy of Technology

Aug 2022 – May 2026

B.E. in Computer Science and Engineering (CGPA: 9.2/10)

Bengaluru, Karnataka

– Key Courses: Data Structures, DBMS, Software Engineering, Machine Learning, Web Technologies, Operating Systems

BGS PU College

Jun 2020 – Mar 2022

PUC (PCMB), Percentage: 94.83%

Bengaluru, Karnataka

Projects and Experience

Malicious SQL Query Detection System: Hybrid CNN + Rule-Based Model

Jul 2025 – Present

- Designed a hybrid CNN and rule-based architecture for real-time SQL injection detection with **95%+ accuracy and 100ms latency**.
- Developed adversarially trained CNN model using temporal data split to reduce overfitting and improve generalization.
- Integrated **SHAP-based model interpretability** for transparent security analytics.
- Built a full-stack web interface (React + FastAPI) for live query monitoring, result visualization, and alert generation.
- **Tech Stack:** Python, TensorFlow, Keras, FastAPI, React.js, MySQL, Pandas, SHAP.

Weather Analytics Dashboard

2025

- Developed a full-stack web application that allows users to search any city and view **real-time weather conditions, hourly trends, and multi-day forecasts** with clean, interactive visuals.
- Implemented secure **Google sign-in**, personalized preferences and favorites, and a **responsive UI with dark mode** for consistent experience across devices.
- Built a **Node/Express API with MongoDB**, integrated the OpenWeatherMap API for live data, and deployed with a **Vercel frontend** connected to a **Railway-hosted backend**.
- **Live Project:** [Weather-Forecast-Dashboard](#)
- **Tech Stack:** Node.js, Express.js, React.js, MongoDB, OpenWeatherMap API, Tailwind CSS, JWT.

Online Bus Ticket Booking and Reservation System

2024

- Developed secure and scalable MERN application for real-time bus reservations with dynamic seat availability and route management.
- Implemented **JWT authentication**, RESTful APIs, and admin dashboard for trip scheduling and fleet management.
- Enhanced UX through interactive seat maps, search filters, and responsive frontend design.
- **Tech Stack:** MongoDB, Express.js, React.js, Node.js, Tailwind CSS, JWT.

Tasks Distributor System with Role-Based Access Control	2024
– Built a role-based task management platform supporting bulk CSV uploads for automatic task assignment among agents. – Secured user access with JWT and bcrypt; added duplicate prevention and conflict-free allocation logic. – Developed real-time dashboard for admins to monitor task distribution, progress, and agent performance. – Tech Stack: Node.js, Express.js, MongoDB, React.js, JWT, bcrypt.	

Excel Data Filtering Platform with Dynamic Querying	2024
– Implemented full-stack system to process and filter Excel datasets dynamically using React and Express backend. – Designed efficient Excel parsing to MongoDB pipeline, achieving sub-second filtering on large datasets. – Created pivot-style table visualization and multi-select dropdown filters with search and “Select All” options. – Tech Stack: React.js, Node.js, Express.js, Python (pandas, openpyxl), MongoDB.	

Publications

Fingerprint Liveness Detection Using Deep Learning	2023
<i>IEEE CIISCA Conference — Co-authored and presented paper</i>	

Certifications and Training

NPTEL — Programming, Data Structures and Algorithms using Python	2023
Infosys Springboard — TechA Cloud Computing using Microsoft Azure Certification	2025
Infosys Springboard — TechA Linux Foundation Certification	2024
Infosys Springboard — TechA Python Programming Foundation Certification	2024
Infosys Springboard — Machine Learning Programming Foundation Certification	2025
Infosys Springboard — Agile Scrum Certification	2025

Achievements & Leadership

- Flipkart GRiD 6.0 (2024):** Developed scalable e-commerce prototype, demonstrating rapid backend design and API efficiency.
- Hack-A-League 3.0 (2025):** Built AI-powered workflow automation prototype under time constraints, showcasing teamwork and innovation.